

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	A.Gopika	Reg. No.:	192521343
Section / Batch:	BTECH - IT	Date:	10/08/2026

Code of Conduct

I A. Gopika (192521343) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Gopika

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Doubly Linked List
5 marks**Q6**Queue
5 marks**Q7**Queue - Deque
5 marks**Q8**Binary Search
5 marks**Q9**

Circular Queue



```
front = (front + 1) % MAX;  
return data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue()  
2 {  
3     if (front==--1)  
4         return -1;  
5     int data=queue[front];  
6     if (front==rear)  
7     {  
8         front=rear--1;  
9     }  
10    else  
11    {  
12        front=(front+1)%MAX;  
13    }  
14    return data;  
15 }
```

Test Input**Output**

Run



Save



Submit Fix

Q7 Stack

5 marks

5 marks

5 marks

5 marks

5 marks

5 marks


```
return stack[top--]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Test Input

```
1 pop(stack)
2     if top== -1:
3 print("underflow")
4     else:
5 return stack[top--]
```

stdin input (optional)



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5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Doubly Linked List

5 marks

Q6

Queue

5 marks

Q7

Queue - Deque

5 marks

Q8

Binary Search

5 marks

Q9

Circular Queue

```
if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
return binarySearch(arr, mid, r, x);  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[],int l,int r, int x)  
2 {  
3     if(l>r)  
4         return-1;  
5     int mid=l+(r-1)/2;  
6     if(arr[mid]==x)  
7         return mid;  
8     if (arr[mid]>x)  
9         return binarysearch(arr,l,mid-1,x);  
10    returnbinary(arr,mid+1,r,x);  
11 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix



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Q2
Circular Queue
5 marks



Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Doubly Linked List
5 marks



Q6
Queue
5 marks



Q7
Queue - Deque
5 marks



Q8
Binary Search



```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data)  
2 {  
3     if(rear==MAX-1)  
4     {  
5         printf("Queue Overflow");  
6         return;  
7     }  
8     rear++;  
9     queue[rear]=data;  
10 }
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix



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Circular Queue
5 marks

Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Doubly Linked List
5 marks



Q6
Queue
5 marks



Q7
Queue - Deque
5 marks



Q8
Binary Search
5 marks



```
del->prev->next = del->next;  
if (del->next != NULL)  
del->next->prev = del->prev;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node**head,Node*del)  
2 {  
3     if(del==NULL)  
4         return;  
5     if (del->prev!=NULL)  
6         del->prev->next=del->next;  
7     else  
8         *head=del->next;  
9     if (del->next!=NULL)  
10        del->next->prev=del->prev;  
11    free(del);  
12 }
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix



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Circular Queue
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Doubly Linked List
5 marks

Q6 ✓
Queue
5 marks

Q7 ✓
Queue - Deque
5 marks

Q8 ✓
Binary Search
5 marks

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop(int stack[])
2 {
3     if(top==--1)
4     {
5         printf("Stack Underflow\n");
6         return-1;
7     }
8     return stack[top--];
9 }
```

▶ Run

☁ Save

🔧 Submit Fix

Test Input

stdin input (optional)

Output





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Circular Queue
5 marks

Q3 ✓
Stack
5 marks

Q4 ✓
Stack
5 marks

Q5 ✓
Doubly Linked List
5 marks

Q6 ✓
Queue
5 marks

Q7 ✓
Queue - Dequeue
5 marks

Q8 ✓
Binary Search
5 marks

Q9 ✓

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(top==MAX-1)
2 {
3     printf("stack Overfull");
4 }
5 else
6 {
7     top++;
8     stack[top]=items;
9 }
10 }
```

Test Input

stdin input (optional)

Output

▶ Run

☁ Save

🔧 Submit Fix



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Q4

Stack

5 marks



Q5

Doubly Linked List

5 marks



Q6

Queue

5 marks



Q7

Queue - Deque

5 marks



Q8

Binary Search

5 marks



Q9

Circular Queue

5 marks



Q10

Stack - Postfix Expression

5 marks



Your Fixed Code

```
1 void insertFront(int data)
2 {
3     if((front==0&&rear==MAX-1) ||
4         (front==rear+1))
5     {
6         printf("Deque Overflow");
7         return;
8     }
9     if(Front==-1)
10    {
11        front=rear=0;
12    }
13    elseif(front==0)
14    {
15        front=MAX-1;
16    }
17    else
18    {
19        front=front-1;
20    }
21    deque[front]=data;
22 }
```

Test Input

stdin input (optional)

Output



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QUESTIONS

Q1

Stack
5 marks



Q2

Circular Queue
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Doubly Linked List
5 marks



Q6

Queue
5 marks



Q1 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int PEEK(int stack[])
2 {
3     if(top==--1)
4     {
5         printf("Stack Underflow\n");
6         return-1;
7     }
8     return stack[top];
9 }
```

Test Input

stdin input (optional)

Output



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Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8

Linked List

5 marks



```
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int DEQUEUE(intQ[])  
2 {  
3     if(front==-1)  
4     {  
5         printf("Underflow\n");  
6         return-1;  
7     }  
8     int item=Q[front];  
9     if(front==rear)  
10        front=rear--1;  
11     else  
12        front++;  
13     return item;  
14 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix



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QUESTIONS

Q1

Binary Search
5 marks



Q2

Stack
5 marks



Q3

Stack
5 marks



Q4

Stack
5 marks



Q5

Stack - Balancing Symbols
5 marks



Q6

Singly Linked List
5 marks



Q7

Stack



Q9 Linked List

5 marks

What is the error in the below code?

new.next = head

head = new

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 struct Node*new=(struct Node*)malloc(sizeof(struct Node));  
2 new->data=value;  
3 new->next=head;  
4 head=new;
```

Test Input

stdin input (optional)

Output

QUESTIONS

- Q1

Binary Search

5 marks

✓
- Q2

Stack

5 marks

✓
- Q3

Stack

5 marks

✓
- Q4

Stack

5 marks

✓
- Q5

Stack - Balancing Symbols

5 marks

✓
- Q6

✓

Q8 Linked List

5 marks

Identify the errors in the following code

```
while current != NULL:
current.next = prev
prev = current
current = current.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while current!=null
2 next=current.next
3 current.next=prev
4 prev=current
5 current=current.next
```

Test Input

stdin input (optional)

Output

 Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to [Settings](#).

- Q7** Stack 5 marks ✓
- Q8** Linked List 5 marks ✓
- Q9** Linked List 5 marks ✓
- Q10** Queue 5 marks ✓

10/10 answered

Your Fixed Code

```
1 #include<stdio.h>
2 int binarysearch(int arr[],int n, int key){
3     int low =0, high = n-2;
4     while(low <= high){
5         int mid = (low + high)/2;
6         if (arr[mid] == key)
7             return mid;
8         else if (arr[mid] < key)
9             low = mid + 1;
10        else
11            high = mid-1;
12    }
13    return -1;
14 }
15 int main(){
16     int n, key;
17     scanf("%d",&n);
18     int arr[n];
```

Test Input

5
10 20 30 40 50
30

Output

 Pending manual review



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5 marks

Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8



What is the issue?

```
temp = head
```

```
while temp.next != NULL:
```

```
print(temp.data)
```

```
temp = temp.next
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 temp=head;
3 while (temp!=NULL)
4 {
5     print(temp.data);
6     temp=temp.next;
7 }
```

Test Input

stdin input (optional)

Output





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5 marks

Q2

Stack

5 marks

**Q3**

Stack

5 marks

**Q4**

Stack

5 marks

**Q5**

Stack - Balancing Symbols

5 marks

**Q6**

Singly Linked List

5 marks

**Q7**

Stack

5 marks

**Q8**

Point out the error:

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop(intstack[])
2 {
3     if(top==1)
4     {
5         printf("Stack underflow\n");
6         return-1;
7     }
8     else
9     {
10         return stack[top--1];
11     }
12 }
```

Test Input

stdin input (optional)

Output





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5 marks

Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8



Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 node*next=NULL;
2 Node*prev=NULL;
3 NODE*CURRENT=head;
4 while(current!=NULL);
5 {
6     next=current->next;
7     current->next=prev;
8     prev=cureent;
9     current=next;
10 }
11 head=prev;
```

Test Input

stdin input (optional)

Output



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5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

In balancing bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
    pop();
} else {
    return 0;
}
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>
2 if((ch==' ' && stack[top]=='(') ||
3    (ch==' ' && stack[top]=='[') ||
4    (ch==' ' && stack[top]=='{'))
5 {
6     pop();
7 }
8 else
9 {
10     return 0;
11 }
12
```

Test Input

stdin input (optional)

Output



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5 marks

Q2

Stack

5 marks



Q3

Stack

5 marks



Q4

Stack

5 marks



Q5

Stack - Balancing Symbols

5 marks



Q6

Singly Linked List

5 marks



Q7

Stack

5 marks



Q8



Identify the errors in the following snippet.

```
if(ch == ')') {  
while(stack[top] != ')') {  
postfix[j++] = pop();  
}  
pop();  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 if(ch==' '){  
3 {  
4     while(stack[top]!='(')  
5     {  
6         postfix[j++]=pop();  
7     }  
8     pop();  
9 }
```

Test Input

stdin input (optional)

Output



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5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack - Balancing Symbols

5 marks

Q6

Singly Linked List

5 marks

Q7

Stack

5 marks

Q8

Linked List

5 marks

Q9

Linked List

```
top = 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include<stdio.h>  
2 int stack[5];  
3 int top;  
4 void init() {  
5     top=-1  
6 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

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🌟 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

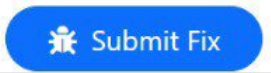
Your Fixed Code

```
1 int pop() {  
2     if (top == -1) {  
3         printf("stack underflow");  
4         return -1;  
5     }  
6     return stack[top--];  
7 }
```

Test Input

stdin input (optional)

Output

 Run  Save  Submit Fix